



Soldering

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Contents of this slide are provided by NTOU VLSI Lab601

Objectives

- Introduction to soldering tools.
- Soldering instructions
- Soldering practice

Soldering Tools Introduction

- Soldering Iron



- Soldering Iron Stand



- Solder



- Desoldering Pump



- Diagonal Cutting Pliers



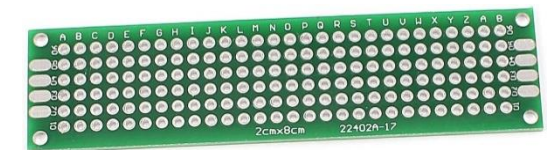
- Needle-nose Pliers



- Wire Strippers



- General-Purpose Printed Circuit Board



Soldering Iron (烙鐵)



- It is used to solder electronic components to the printed circuit board (PCB)(印刷電路板).
- A 30 W soldering iron has a temperature of about $300^{\circ}\text{C} \sim 400^{\circ}\text{C}$ is sufficient for general soldering.
- The higher the wattage (W), the higher the temperature of the soldering iron.
- **Safety Notice**
 - Always handle the soldering iron carefully, as it is very HOT.
 - If it comes in contact with the skin, immediately rinse it with cold water to prevent burns.

Soldering Iron Stand (烙鐵架)



- It is made of metal, designed to hold the soldering iron.
- It comes with a cleaning sponge to clean the soldering iron tip before use.
- The sponge should be slightly wetted when used.
- Before soldering, it is highly suggested to clean the soldering iron tip on the sponge.

Solder (焊錫)



- Use high-quality solder with at least 60% tin content. The remaining 40% consists of lead and other impurities.
- The melting point of ordinary solder is about 180°C .
- There are flux (助焊劑) in the core of the solder to make the soldering much easier.

Desoldering Pump (吸錫器)



- It is used to collect the melted solder when removing or replacing electronic components.
- Desoldering Process:
 1. Push the plunger of the desoldering pump down.
 2. Heat the solder joint with the soldering iron until the solder melts.
 3. Remove the soldering iron and place the desoldering pump near the solder joint immediately.
 4. Press the button to suck the melted solder.
 5. Repeat if necessary.

Diagonal Cutting Pliers (斜口鉗)



- It is primarily used for cutting wires or excess component leads.
- It can also be used to strip the insulation from wires.

Needle-nose Pliers (尖嘴鉗)

Figure source:

<https://hocom.tw/h/ProductDetail?key=893587258333&productId=65957>



- It is used to bend wires and component leads.
- It can be regarded as a pair of cutting pliers as it also has a cutting edge.
- It has long and narrow tips to work in small spaces.
- It is often used to place or remove ICs and wires.

Wire Strippers (剝線鉗)

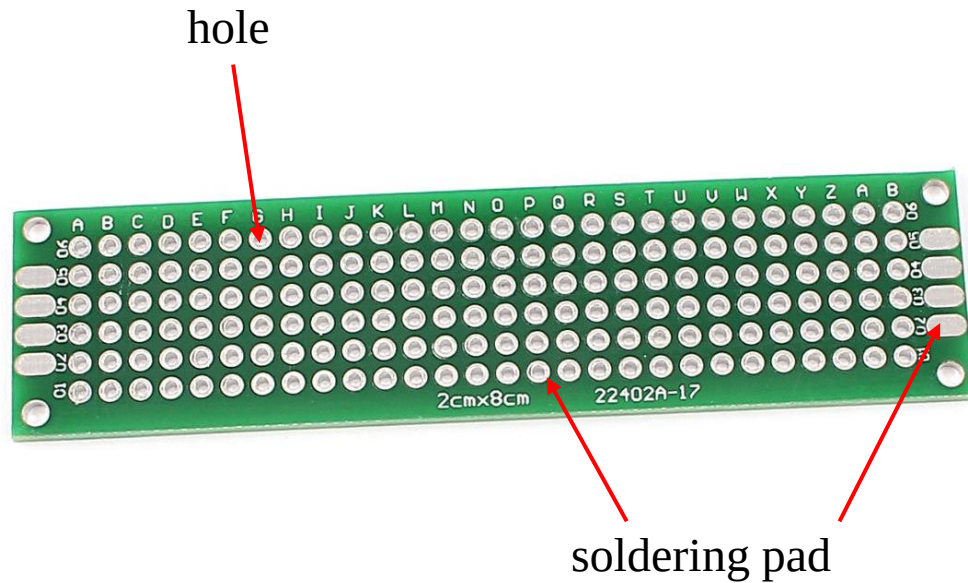


- It is used to remove the plastic insulation from wires.
- It has various sizes of notches to fit different wire gauges.
- To use, position the wire in the correct notch, squeeze the handles to strip the insulation, and pull it off.

General-Purpose Printed Circuit Board (PCB)

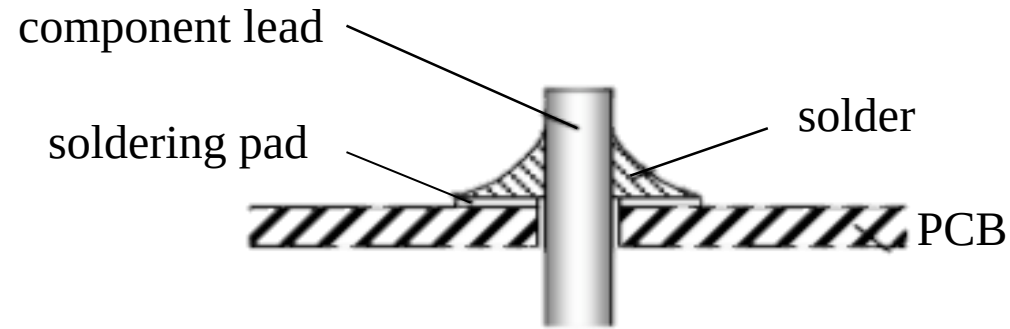
(萬用板/洞洞板)

Figure source:
<https://shop.mirotek.com.tw/shop/600060/>



- A general-purpose PCB, also known as a prototype board, is commonly used for electronic designs.
- It is filled with standard IC spacing and independent soldering pads, and the whole board is covered with small holes.

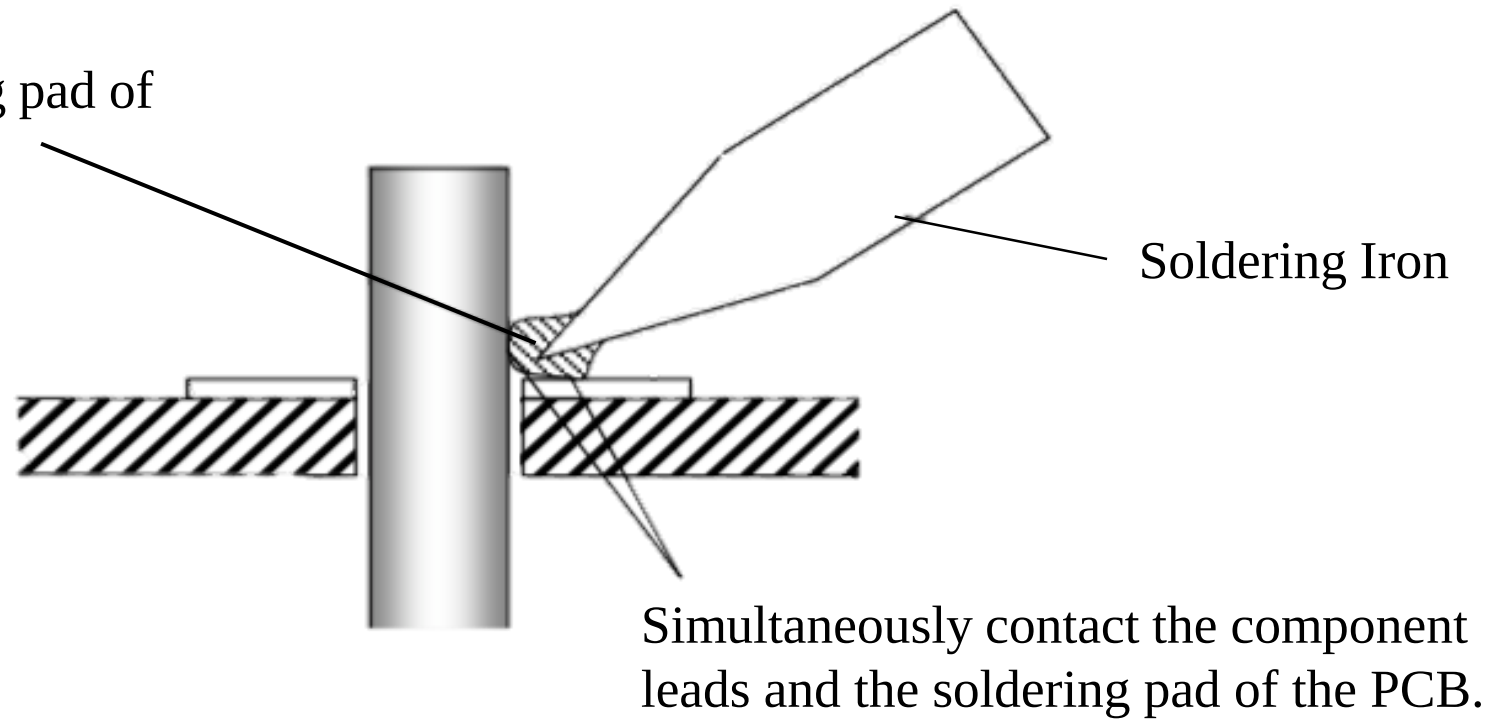
Soldering Explanation



- Soldering involves using solder to joint two pieces together.
- Good soldering requires that the solder flows into the metal surfaces, forming a good electrical contact.
- The metal surfaces to be soldered must be clean. If the metal surface is oxidized, successful soldering will not occur.
- Soldering temperature is around 180°C .
- Solder usually contains flux to prevent oxidation at high temperatures, leaving harmless yellow-brown flux residue after soldering.

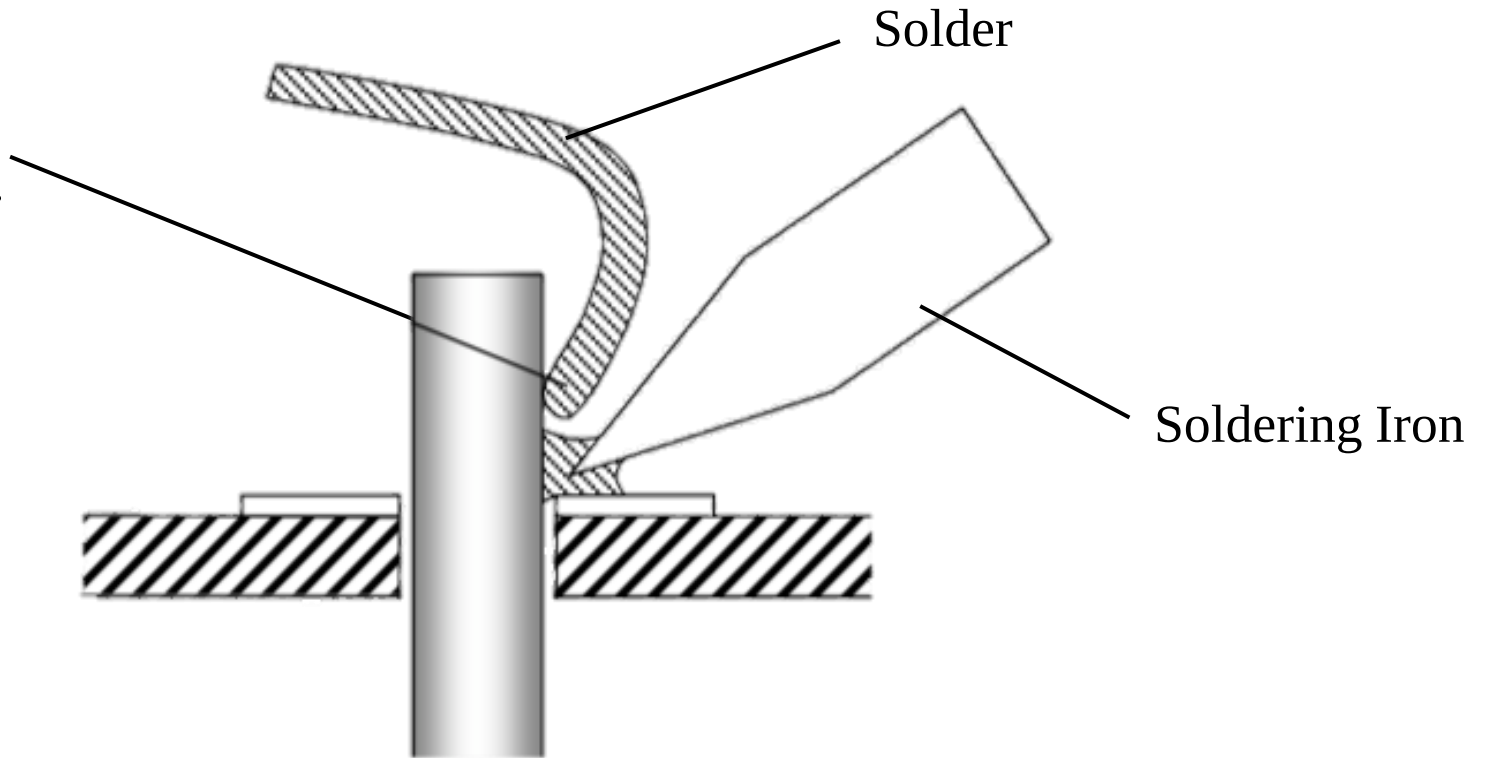
Soldering Steps – 1

The tip of the soldering iron should have a small amount of solder to increase the contact area, allowing the heat to be effectively transferred to both the component leads and the soldering pad of the PCB.



Soldering Steps – 2

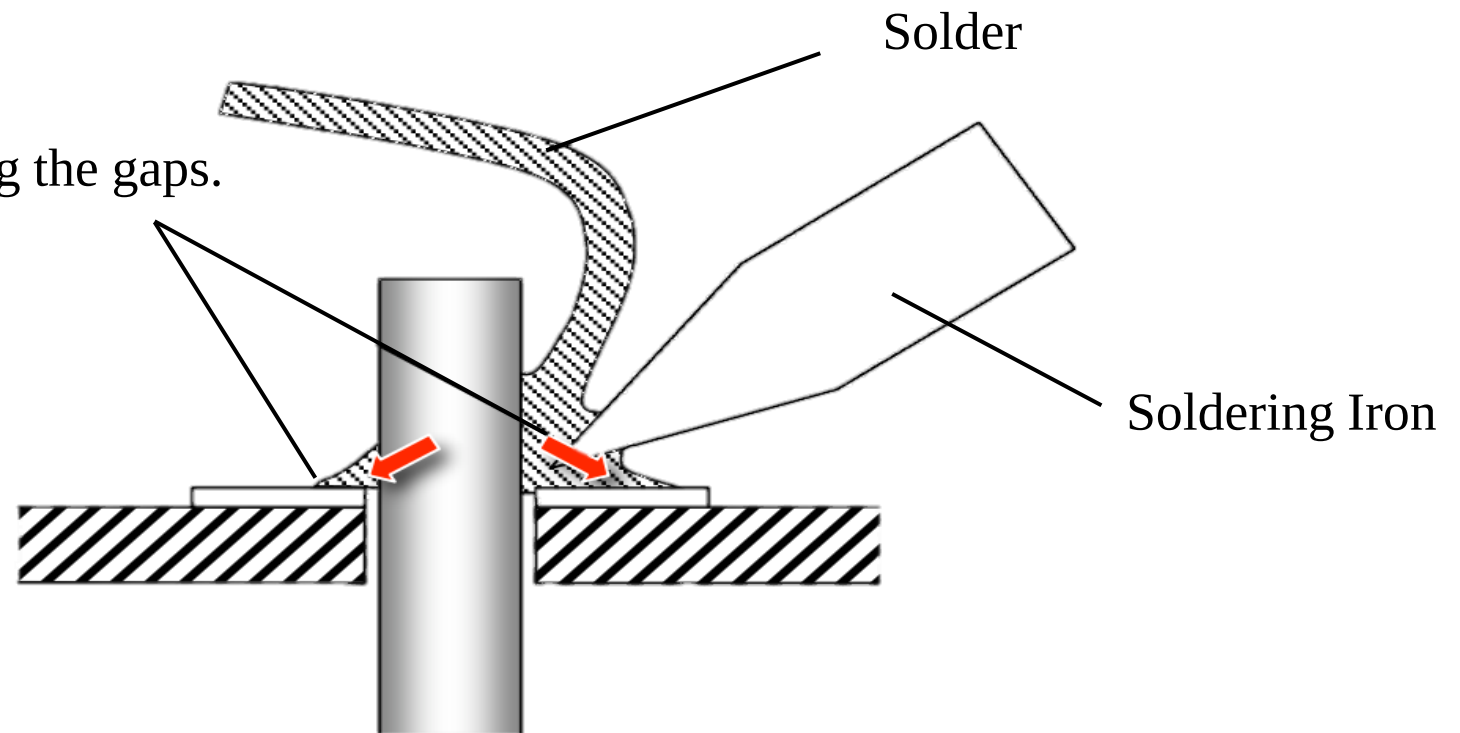
Add solder to the component leads or the soldering pad (not to the tip of the soldering iron).



Soldering Steps – 3

- The solder will flow naturally due to the attraction between the metal atoms, adhering to the component lead and the soldering pad of the PCB.
- If the solder does not flow naturally, it may be due to an unclean metal surface or insufficient contact area from the soldering iron.

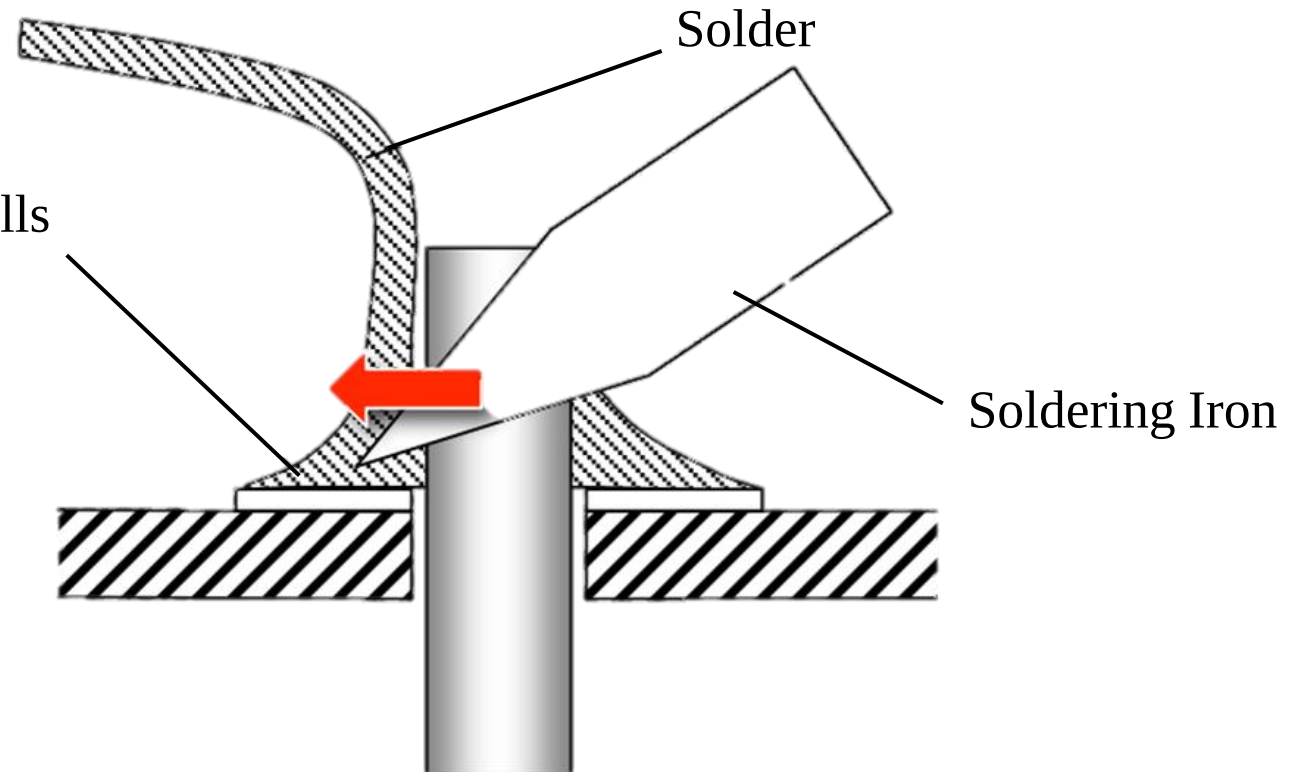
The solder will flow naturally, filling the gaps.



Soldering Steps – 4

- Sometimes the solder does not spread evenly. In that case, move the soldering iron and the solder at the same time to ensure even distribution of solder.

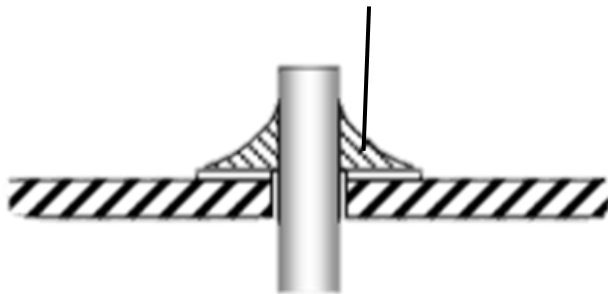
Move the solder and soldering iron to ensure the appropriate amount of solder fills the joint.



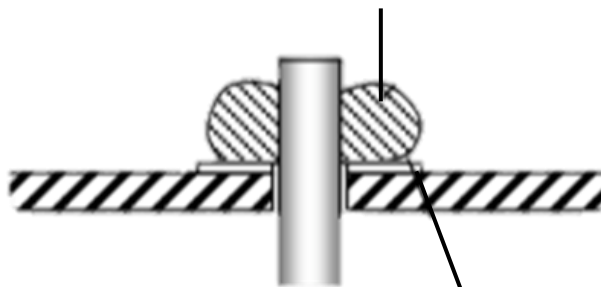
Solder Joint Examples

- If the soldering iron stays on the joint for more than 3 seconds, the component or PCB may be damaged.
- Prolonged heating can also oxidize the solder, resulting in poor solder joints.
- If the soldering does not work, you can use a desoldering pump to remove the solder.
- If the solder has oxidized, add some new solder to restore the shine.

Good solder joint (焊點)

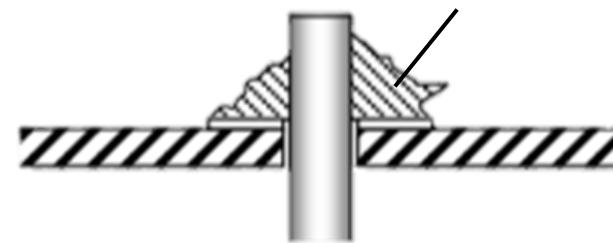


If the solder joint is in the shape of a ball, it indicates that the solder did not flow into the soldering pad.

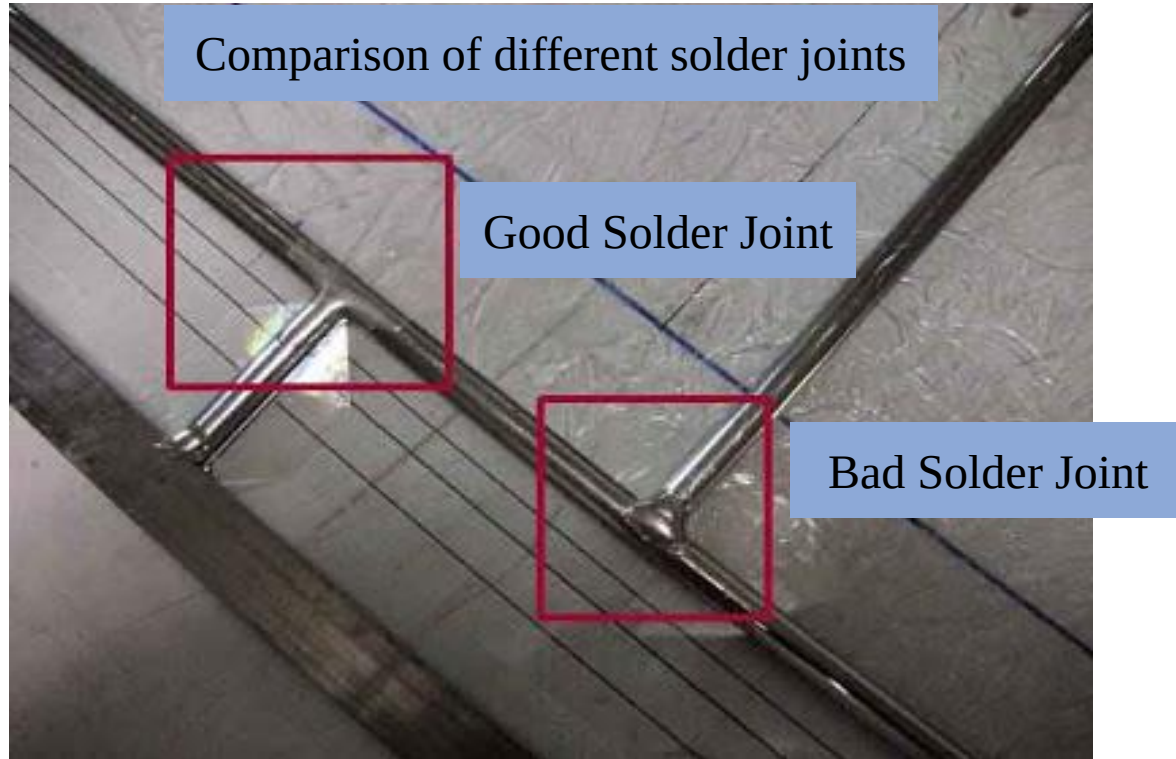


The soldering pad was not clean enough.

When excessive heating, the solder surface becomes irregular and loses its shine due to overheating.



Solder Joint Examples



Soldering Tutorial

- <https://www.youtube.com/watch?v=UUIHBjsaMeM>
- <https://www.youtube.com/watch?v=VCyuhqx5eSg>
- <https://www.youtube.com/shorts/D5IvlJVqa7M>
- <https://www.youtube.com/shorts/b4l-gZmdUR8>
- <https://www.youtube.com/shorts/g2QraxxLKUs>

Soldering Practice

- Please use enameled wires (漆包線) to solder a square pyramid.
- Basic requirements (MUST MEET ALL):
 - All edges should be 4 cm in length.
 - There are 5 joints, 4 with 3 wires each, and 1 with 4 wires.
- HINT: Remove the enamel at the places where you want to solder.
- Grading Criteria:
 - Basic requirements met: 70 points
 - Difficulty of the product: 10 points
 - Quality of the joints and neatness: 10 points
 - Durability of the product: 10 points
- Bonus: Modify the overall shape while meeting the above requirements.

